

Prediction of Concrete Compressive Strength using various Regression models and XAI

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Abstract

Concrete compressive strength (CCS) prediction is a fundamental performance indicator in construction engineering, valid to both standard and high-performance concrete structures. Because physical development processes are so complex, CCS is influenced by a broad spectrum of circumstances, making quantification extremely difficult. This study employs a dataset from the current literature with 1030 data points from 9 input variables, ingredients include cement, blast furnace slag, water, superplasticizer, and fine aggregate, and concrete age, to model and estimate the sensitivity of CCS. In this research, Polynomial Regression was discovered to be the most promising regression model for predicting outcomes. Each model provides distinct insights into the data. Polynomial Regression is also used because of its superior capacity to capture non-linear correlations between input factors and CCS. The incorporation of these regression models into the ensemble technique not only improves forecast accuracy, but also gives a thorough framework for assessing the CCS. Understanding the parameters that determine CCS is critical for the construction business. This study uses sophisticated Explainable artificial intelligence (AI) approaches, notably Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP), to demystify the predictive models used to estimate concrete compressive strength. By combining these explainable AI methodologies with ML models, we want to provide unambiguous insights into the influence of each input variable. The study's outcomes offer significant perspectives on refining concrete mix designs for increased compressive strength. Understanding the impacts of lime addition and aggregate shap allows engineers and practitioners to adjust concrete formulations to specific performance needs while guaranteeing cost-effectiveness and sustainability in construction projects.

Introduction

Concrete is a heterogeneous composite material made of several elements such as cement, water, blast furnace slag, superplasticizer, age, fly-ash, fine and coarse aggregates which plays a major part in construction. The CCS, specimen being tested is also heavily influenced by its size and shape. Because there is more surface area to sustain the force, the CCS sample decreases with increasing size. Cylindrical samples usually have higher strength than cubical or irregular-shaped ones. One of its key features is CCS, which has a direct impact on structural performance. Reliable prediction of CCS is necessary for maintaining the safety and durability of concrete buildings. CCS represents its capacity to sustain axial loads .Concrete compressive strength testing is critical for construction quality control. This property is evaluated using a wide range of ways, the compression test being the most common. This approach includes progressively increasing pressure on a concrete cylinder in a machine until it fractures. The cylinder's compressive strength is then measured by the largest load it can sustain without breaking.

In recent years, machine learning (ML) techniques have emerged as powerful tools for predicting CCS, leveraging vast datasets to uncover complex relationships between concrete components and their compressive strength. Among these techniques, regression models have shown significant promise due to their ability to model the non-linear interactions between variables. This research explores the application of various regression models, including Linear Regression, Ridge Regression, LASSO, and tree-based models like Decision Trees, Random Forests, and XGBoost, to predict CCS.

Moreover, the integration of Explainable Artificial Intelligence (XAI) techniques, such as SHapley Additive exPlanations (SHAP), provides valuable insights into the contribution of each input variable to the model's predictions. This transparency is crucial for civil engineers, enabling them to make informed decisions regarding concrete mix design and optimize construction practices. By combining regression models with XAI, this study aims to enhance the reliability and interpretability of CCS predictions, ultimately contributing to safer and more efficient construction methodologies.

The next is how the paper's organizational framework plays out : Section 2 It offers the background through a relevant literature review. Following next, Section 3 explains the methodologies chosen, that provides comprehensive perception of the research approach. Section 4 provides a full overview of the outcomes and discussions, highlighting major findings and their consequences. Section 5 provides decisive observations and overall remarks, bringing this research to a conclusion.

Related Work

In Güçlüer et.al.[1] The study compares ML algorithms like ANN, DT, SVM and LR algorithms, for predicting CCS, finding DT to yield the highest precision with an R2 of 0.86 and a MAE of 2.59, suggesting its suitability for efficient estimation. In Ahmad, A. et.al[2] This study explores the application of supervised ML techniques, including DT, ANN, bagging, and gradient boosting, to predict CCS at high temperatures. Evaluation metrics indicate strong correlations and improved performance with ensemble algorithms, highlighting their efficacy in enhancing predictive accuracy and mitigating the effect of high temperatures on concrete properties. In Imran et.al[3] Multivariate polynomial regression, which performed better in the context of estimation accuracy than linear regression and SVM models, was used in the study to construct a estimation model for the CCS mixtures. Through the optimization of eco-friendly CCS, this model can aid in lowering the environmental effect of concrete production. In Rajakurunakaran et.al [4] Using a laboratory dataset of 99 samples, ML regression models were created to forecast SCC compressive strength. After comparing several approaches, Random Forest was the best accurate predictor. SCC compression strength can be accurately and efficiently predicted by the RF model. In Naser et.al[5] In order to estimate the CCS, this study creates a ML model utilizing Multivariate Adaptive Regression Splines (MARS). Considering the ratio of determination (R2) of 0.889 and a Root Mean Squared Error (RMSE) of 4.110 MPa, the MARS model outperforms than the SVM and RF regression algorithms. The approach could be useful to civil engineers for creating long-lasting infrastructure designs. In Tafese et.al[6] In this study, a machine learning model for concurrently predicting chlorine migration and compressive strength in concrete is developed. It determines that gradient boosting is the top performer among decision tree-based ensemble techniques as bagging, random forest, AdaBoost, and XGBoost. The best optimal path is Scenario 2, which has optimal hyperparameters. The representation can be customised by adding more varied concrete qualities, even though it already demonstrates excellent generalization ability and potential for concrete design. In Ghunimat et.al[7] This study predicts CCS with fly ash and GGBFS using supervised learning and neural networks. Using variables such as cement, water, aggregates, superplasticizer, and maturity age as inputs, it applies the MLP, RFR, and KNN regression techniques with Python. RFR and MLP outperformed KNN with regards to precision and error reduction. The model that was least impacted by data splitting was determined to be RFR. In Ahmed et.al[8] Utilising soft computing models, the compressive strength of ground granulated blast furnace slag (GGBFS) concrete is predicted in this piece of writing. Water content, temperature, w/b ratio, GGBFS/b ratio, fine and coarse aggregates, and superplasticizer dosage are the seven factors which are utilised with data from 268 samples. With a dependable correlation of 0.9765 and 0.9522, respectively, the SVR-PSO and SVR-GWO models yield the greatest outcomes.In Jibril et.al[9] Through the prediction of compressive strength, this research attempts to optimize high-performance concrete compositions. The input variables, such as cement, cement strength, superplasticizer, fly ash, etc., are analyzed using FFNN, ENN, SVM, and MLR. The output is the CCS after 28 days. The models have strong predictive resilience; FFNN-M4, ENN-M4, SVM-M4, and MLR-M4 exhibit the greatest results, with R2 values of 0.9950, 0.9853, 0.9736, and 0.9678, in that order? High-performance concrete's compressive strength can be accurately and successfully predicted using the suggested approach.In Ahmad et.al[10] This work aims on the manufacture of concrete and mortar as a particular example of the sustainability dilemma facing the construction sector. It investigates replacing cement with used glass to lessen the impact on the environment. To be able to predict the CCS after adding waste glass powder, the study suggests three models: linear regression, nonlinear regression, and artificial neural network. Compressive strength-related factors were considered, including waste glass content and curing duration. The ANN model outperformed the others, based on the results, and these models can predict the CCS of typical concretes with different waste glass dosages.

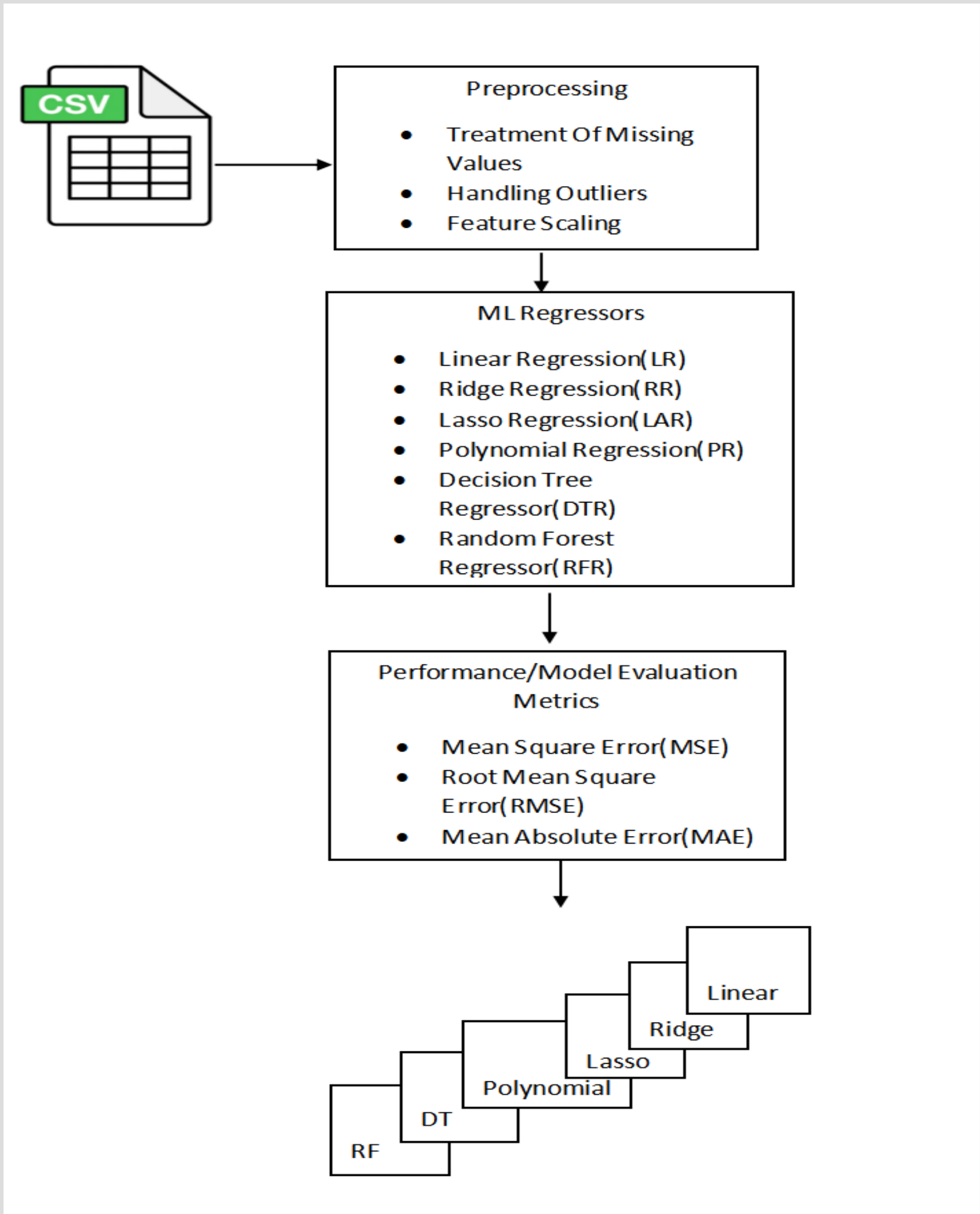
In Candelaria et.al[11] This study offers parameters for estimating the CCS in marine situations. It tests physical and design factors using cylindrical samples of changing saturation and salinity. Machine learning methods, notably ANN, forecast strength successfully, with a focus on P-wave velocity, electrical resistivity, and the water-to-binder ratio for practical assessment. In Li et.al[12] The research provides a bibliometric analysis of CCS prediction, employing 3135 articles to chart the field's evolution. The article divides prediction approaches into classical and machine-learning, emphasizing the GBRT algorithm's high accuracy (R2) of 0.92 and RMSE of 4.7 MPa based on 1030 datasets. In Kumar et.al[13] In this work AI-driven models such as optimized GPR, SVMR, and EL provide precise forecasts for the compressive strength of lightweight concrete, exceeding older empirical approaches. The optimized GPR model performs admirably with a (R = 0.9803), demonstrating that it is both cost-effective and efficient for industrial use. In Mansouri et.al[14]This research describes a ML approach for estimating the compressive strength of eco-friendly geopolymer concrete. It integrates CatBoost, additional trees, and gradient boosting regressors, and improved by 13% with a hybrid model, displaying great prediction accuracy across 147 samples. In Varma et.al[15]The study investigates ML approaches, GBM and LGBM, for predicting CCS utilizing eight factors. The LGBM model surpasses GBM with a (R2) of 0.916, showing that cement content, fine aggregate, and water content are critical predictors. In Shen et.al[16]This research predicts the compressive strength of UHSC using various techniques including as XGBoost, AdaBoost, and Bagging. XGBoost excels with a (R2) of 0.90. SHAP analysis identifies curing time as the most relevant element for strength prediction. In Malhotra et.al[17]In this work of ML approaches for forecasting concrete strength, fifteen alternative algorithms were examined. The Support Vector Regressor (SVR) achieved the greatest accuracy rate of 88.18%, making it very successful in handling continuous outcomes and complex, nonlinear interactions between variables. In Paudel et.al[18]The study compares ML algorithms for predicting CCS with fly ash and finds that XGBoost outperforms the others. It focuses on the concrete's age, cement, and water as crucial components, providing a less expensive alternative to rigorous lab testing. In Nguyen et.al[19]The study demonstrates how SVR, MLP, GBR, and XGBoost ML can be utilised to forecast HPC strengths. GBR and XGBoost models excel at predicted accuracy and efficiency when hyperparameter adjustment and average imputation for absent data are used. In Sapkota et.al[20]The study uses advanced ML techniques, particularly RF and CatBoost, to estimate CCS with accuracies of 89.3% and 95.8%, respectively. CatBoost outperforms expectations, demonstrating ML's potential for improving sustainable infrastructure. Ruchitha et. al. [21] used several ML regressors to determine the cement strength using 16 features. Mandre et al. [22] employed Linear, Ridge and Lasso regressors to determine the area of forest fire.

Methodology

A. Dataset Description

The purpose of this investigation is to predict the CCS using several ML methods. The dataset is drawn from the Kaggle Database, specifically the “Concrete Compressive Strength” dataset. The dataset comprises of 1030 samples and 9 attributes. The target variable is the concrete mixture's compressive strength The features that are included in this dataset are superplasticizer, blast furnace slag, water, cement, coarse aggregate, fly ash, fine aggregate (measured in kg/m³) and age. All the 9 attributes are numerical in nature with integer and float datatype. While performing Exploratory Data Analysis It occurred that there were some missing values in two features namely slag and coarse aggregate.

B. Flowchart Description:



Mean imputation was utilised in this dataset to impute the values that were absent for two features, namely coarse aggregate and slag. Drawing a box plot on numerical attributes yields information on outliers. It was noticed that the dataset contained no outliers. Since coarse and fine aggregate were the dataset's dominant features, Min-Max Scaler was used. There was utilisation of the single regressors. Specifically, Random Forest Regressor, Polynomial Regression, Decision Regressor, Ridge Regression, Lasso Regression, and Linear Regression. Ensemble learning strategy such as RFR is used to build a considerable amount of decision trees during training, and the outcome is the median estimate(regression) of each individual tree. Curved correlations between predictors and responses can occur when a polynomial function is fitted to the information using polynomial regression. Decision trees, which segment the area of features into regions, are accustomed to forecast the average response within each area. For the purpose To avoid overfitting, the LR cost function in ridge regression includes a penalty component. Lasso regression incorporates a penalty component notwithstanding that it utilizes the whole value of the coefficients. The simplest regression model fits predictors and responses in a linear fashion. One common metric to utilize is the MSE when assessing how well does a model based on regression fit the data. The number being measured indicates the average of all values squared difference b/w anticipated values and the goal values. RMSE quantifies the average deviation between a statistical model's projected values and the dataset's actual values. The MAE metric is applied to calculate regression model performance. The standard error magnitude is measured comparing expected and actual values. Each type has its merits and limitations. Makes use of a RFR to combine many decision tree regressors into one, dependable, control non-linear connections, and helps prevent overfitting. PR is utilised to represent relationships as polynomial functions. helpful in obtaining non-linear patterns. The Decision Tree Regressor forecasts continuous target values by creating a framework that resembles a flowchart. an element of group approaches like random forests. Reducing coefficients using Ridge Regression helps prevent overfitting. Lasso Regression encourages sparsity and manages high-dimensional data. LR is the name for simple linear equation modelling. used a lot in predictive modelling.

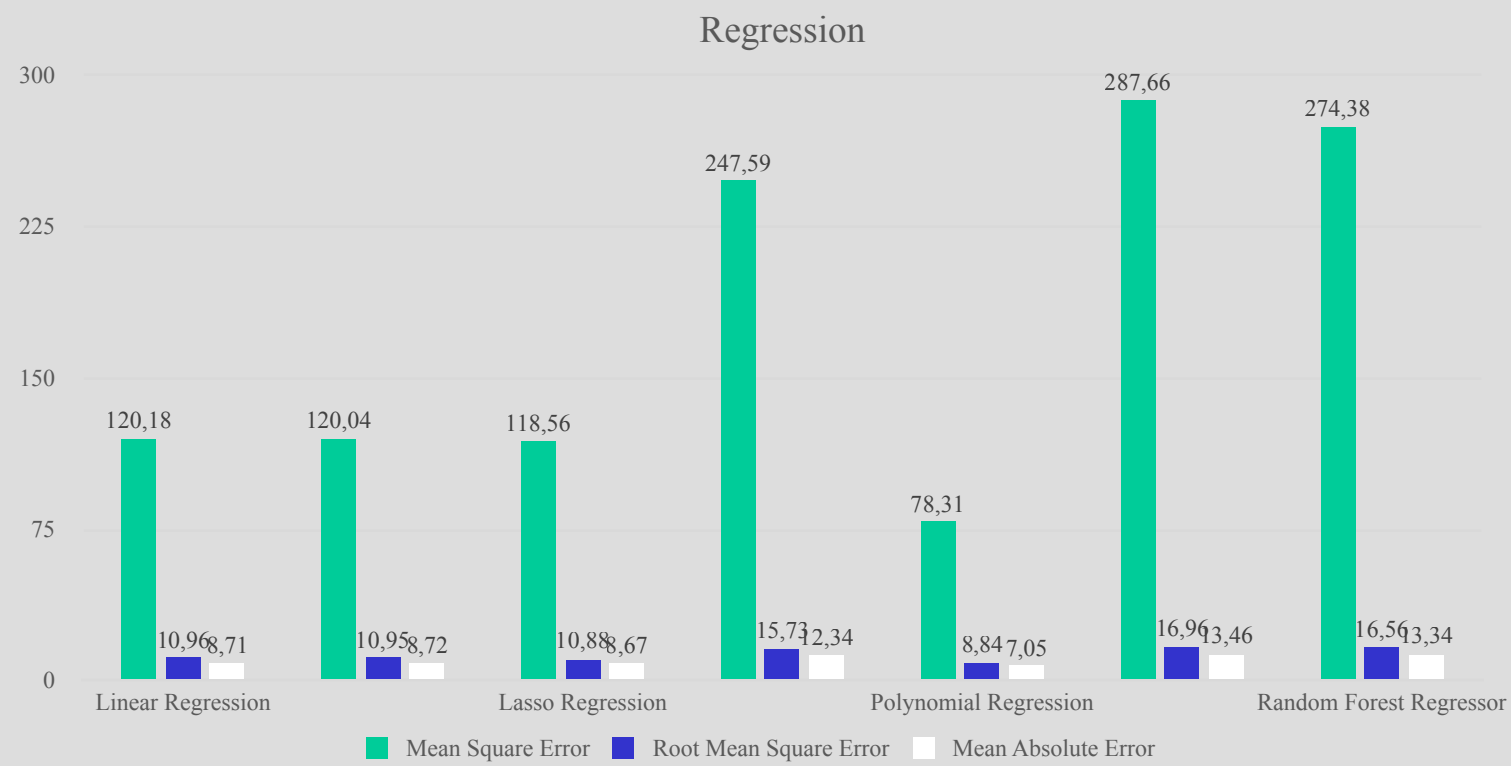
Theoretical Modelling

Here's a brief theoretical overview of each regression model:

- Linear Regression: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$
- Ridge Regression: $Loss = i = 1 \sum n (y_i - \hat{y}_i)^2 + \lambda_j = 1 \sum p \beta_j^2$
- Lasso Regression: $Loss = i = 1 \sum n (y_i - \hat{y}_i)^2 + \lambda_j = 1 \sum p |\beta_j|$
- Elastic Net Regression: $Loss = i = 1 \sum n (y_i - \hat{y}_i)^2 + \lambda_1 j = 1 \sum p |\beta_j| + \lambda_2 j = 1 \sum p \beta_j^2$
- Polynomial Regression: $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_n X^n + \epsilon$
- Decision Tree Regressor: $Prediction = N1 i = 1 \sum N y_i$ (for each leaf node)
- Random Forest Regressor: $Prediction = T1 i = 1 \sum T Treet(X)$

Results and Discussion

In this study, we aim to predict concrete compressive strength using various regression models and Explainable AI (XAI) techniques. We begin by collecting a dataset with features such as cement content, water-cement ratio, age of concrete, fine aggregate, and coarse aggregate. After preprocessing the data to handle missing values and normalize features, we split it into training and testing sets. We then implement several regression models, including Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression, Polynomial Regression, Decision Tree Regressor, and Random Forest Regressor. Each model is trained on the training set and evaluated on the testing set using metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). To enhance model interpretability, we apply XAI techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). These techniques help us identify the most influential features and provide insights into the model's decision-making process. Our results show that models like the Polynomial Regression perform well in predicting compressive strength, with SHAP values highlighting the importance of features like cement content and water-cement ratio. This approach not only improves prediction accuracy but also offers valuable insights for optimizing concrete mix designs and enhancing the quality and durability of concrete structures.



In this research, Polynomial Regression emerged as the best ML model for predicting Concrete Compressive Strength (CCS), achieving the lowest error rates (MSE: 78.31, RMSE: 8.84, MAE: 7.05). SHAP and LIME analyses revealed that cement and fine aggregate significantly influence CCS, with age having the highest model impact.

Regression	Mean Square Error	Root Mean Square Error	Mean Absolute Error
Linear Regression	120.18	10.96	8.71
Ridge Regression	120.04	10.95	8.72
Lasso Regression	118.56	10.88	8.67
Elastic Net Regression	247.59	15.73	12.34
Polynomial Regression	78.31	8.84	7.05
Decision Tree Regressor	287.66	16.96	13.46
Random Forest Regressor	274.38	16.56	13.34

Conclusion

The study explores the development of machine learning (ML) regression models to forecast concrete compressive strength (CCS) based on mixture composition and curing age. These predictive models are crucial in improving construction practices by accelerating processes and optimizing resource allocation. By leveraging ML, researchers can assess complex relationships between multiple factors, such as aggregate type, water-to-cement ratio, and cement consumption, which traditionally required resource-intensive laboratory testing. The application of ML techniques, including regularization methods like Lasso and Ridge, has demonstrated the potential for robust and accurate CCS predictions. Feature engineering, data quality, and hyper-parameter tuning are pivotal in refining these models. Moreover, ML enables the exploration of sustainable practices, such as incorporating alternative materials like fly ash and recycled aggregates, thus reducing the environmental impact of concrete production. As research progresses, ML regressors are expected to yield even more precise and reliable predictions, paving the way for safer, more efficient, and eco-friendly construction methods.

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